



AI POWERED EV BATTERY SYSTEM

INTELLIGENT BATTERY HEALTH DASHBOARD

DARK

LIGHT

SOH STATUS

--%

CHARGE LEVEL

--%

TEMPERATURE

--°C

SYSTEM



STANDBY

SOH ESTIMATION

RL OPTIMIZATION

MODEL COMPARISON

STATE OF HEALTH ESTIMATION

CNN Model



▶ START SYSTEM

UPLOAD BATTERY DATA

Click to select a single CSV file

PROGRESS

0 / 0

MODEL MAPE

4.09%

BATTERY STATUS

--

Predicted SOH (%) Actual SOH (%)

1.0

0.9



AI POWERED EV BATTERY SYSTEM

INTELLIGENT BATTERY HEALTH DASHBOARD

DARK

LIGHT

SOH STATUS

--%

CHARGE LEVEL

--%

TEMPERATURE

--°C

SYSTEM



STANDBY

SOH ESTIMATION

RL OPTIMIZATION

MODEL COMPARISON

REINFORCEMENT LEARNING LOGS

Q-Learning (Baseline) ▼

RECOMMENDED ACTION

--

CUMULATIVE REWARD

0.0

ACTION EFFICIENCY

--%

Project: On-Time Battery Health Estimation and Optimization in Electric Vehicles

Team: Md. Shiban Alam, Syed Sarafeena Ali, Purbasha Roy, Dindira Sain

Mentor: Mrs. Amrita Bhattacharya | **Institution:** STCET - MAKAUT

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AI POWERED EV BATTERY SYSTEM

INTELLIGENT BATTERY HEALTH DASHBOARD

DARK

LIGHT

SOH STATUS

--%

CHARGE LEVEL

--%

TEMPERATURE

--°C

SYSTEM



STANDBY

SOH ESTIMATION

RL OPTIMIZATION

MODEL COMPARISON

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CNN Model



▶ START SYSTEM

UPLOAD BATTERY DATA

Click to select a single CSV file

PROGRESS

0 / 0

MODEL MAPE

4.09%

BATTERY STATUS

--

Predicted SOH (%) Actual SOH (%)

1.0

0.9

 Quick access  This PC  3D Objects  Desktop  Documents  Downloads  Music  Pictures  Videos  Local Disk (C:)  Network	Name	Date modified	Type
	 nasa_test	01-05-2026 16:08	Microsoft Excel
	 nasa_train	01-05-2026 16:08	Microsoft Excel
	 nasa_val	01-05-2026 16:09	Microsoft Excel
	< >		

🌟 Quick access 💻 This PC 📦 3D Objects 🖥 Desktop 📄 Documents ⬇ Downloads 🎵 Music 🖼 Pictures 🎬 Videos 💾 Local Disk (C:) 🌐 Network	Name	Date modified	Type
	📄 nasa_test	01-05-2026 16:08	Microsoft Excel
	📄 nasa_train	01-05-2026 16:08	Microsoft Excel
	📄 nasa_val	01-05-2026 16:09	Microsoft Excel



AI POWERED EV BATTERY SYSTEM

INTELLIGENT BATTERY HEALTH DASHBOARD

DARK

LIGHT

SOH STATUS

--%

CHARGE LEVEL

--%

TEMPERATURE

--°C

SYSTEM



STANDBY

SOH ESTIMATION

RL OPTIMIZATION

MODEL COMPARISON

STATE OF HEALTH ESTIMATION

CNN Model



▶ START SYSTEM

UPLOAD BATTERY DATA

Click to select a single CSV file

Loaded: nasa_test.csv | Total Cycles Found: 73

PROGRESS

0 / 0

MODEL MAPE

4.09%

BATTERY STATUS

--

Predicted SOH (%) Actual SOH (%)

1.0



AI POWERED EV BATTERY SYSTEM

INTELLIGENT BATTERY HEALTH DASHBOARD

DARK

LIGHT

SOH STATUS

--%

CHARGE LEVEL

--%

TEMPERATURE

--°C

SYSTEM



STANDBY

SOH ESTIMATION

RL OPTIMIZATION

MODEL COMPARISON

STATE OF HEALTH ESTIMATION

CNN Model



▶ START SYSTEM

UPLOAD BATTERY DATA

Click to select a single CSV file

Loaded: nasa_test.csv | Total Cycles Found: 73

PROGRESS

0 / 0

MODEL MAPE

4.09%

BATTERY STATUS

--

Predicted SOH (%) Actual SOH (%)

1.0



AI POWERED EV BATTERY SYSTEM

INTELLIGENT BATTERY HEALTH DASHBOARD

DARK

LIGHT

SOH STATUS

--%

CHARGE LEVEL

--%

TEMPERATURE

--°C

SYSTEM



ACTIVE

SOH ESTIMATION

RL OPTIMIZATION

MODEL COMPARISON

STATE OF HEALTH ESTIMATION

CNN Model



STOP

UPLOAD BATTERY DATA

Click to select a single CSV file

Live SOH Estimation Started...

PROGRESS

0 / 0

MODEL MAPE

4.09%

BATTERY STATUS

--

Predicted SOH (%) Actual SOH (%)

1.0



AI POWERED EV BATTERY SYSTEM

INTELLIGENT BATTERY HEALTH DASHBOARD

DARK

LIGHT

SOH STATUS

100.0%

CHARGE LEVEL

77.6%

TEMPERATURE

35.6°C

SYSTEM



ACTIVE

SOH ESTIMATION

RL OPTIMIZATION

MODEL COMPARISON

STATE OF HEALTH ESTIMATION

CNN Model

STOP

UPLOAD BATTERY DATA

Click to select a single CSV file

Live SOH Estimation Started...

PROGRESS

3 / 73

MODEL MAPE

4.09%

BATTERY STATUS

Healthy

Predicted SOH (%) Actual SOH (%)

100

PROGRESS

13 / 73

MODEL MAPE

4.09%

BATTERY STATUS

Healthy



Project: On-Time Battery Health Estimation and Optimization in Electric Vehicles

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AI POWERED EV BATTERY SYSTEM

INTELLIGENT BATTERY HEALTH DASHBOARD

DARK

LIGHT

SOH STATUS

100.0%

CHARGE LEVEL

62.4%

TEMPERATURE

39.4°C

SYSTEM



ACTIVE



SOH ESTIMATION



RL OPTIMIZATION



MODEL COMPARISON

REINFORCEMENT LEARNING LOGS

Q-Learning (Baseline)



RECOMMENDED ACTION

Moderate

CUMULATIVE REWARD

44.0

ACTION EFFICIENCY

85.0%

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PROGRESS

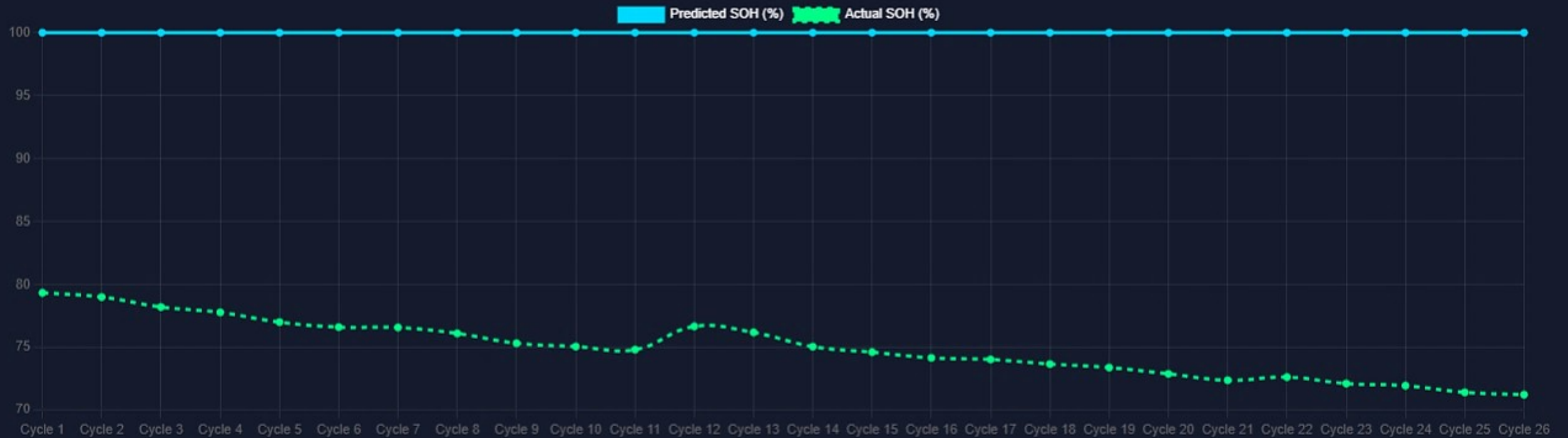
26 / 73

MODEL MAPE

4.09%

BATTERY STATUS

Healthy



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AI POWERED EV BATTERY SYSTEM

INTELLIGENT BATTERY HEALTH DASHBOARD

DARK

LIGHT

SOH STATUS

100.0%

CHARGE LEVEL

53.6%

TEMPERATURE

41.6°C

SYSTEM



ACTIVE

SOH ESTIMATION

RL OPTIMIZATION

MODEL COMPARISON

STATE OF HEALTH ESTIMATION

CNN Model



STOP

UPLOAD BATTERY DATA

Click to select a single CSV file

Live SOH Estimation Started...

PROGRESS

33 / 73

MODEL MAPE

4.09%

BATTERY STATUS

Healthy

100

Predicted SOH (%) Actual SOH (%)

PROGRESS

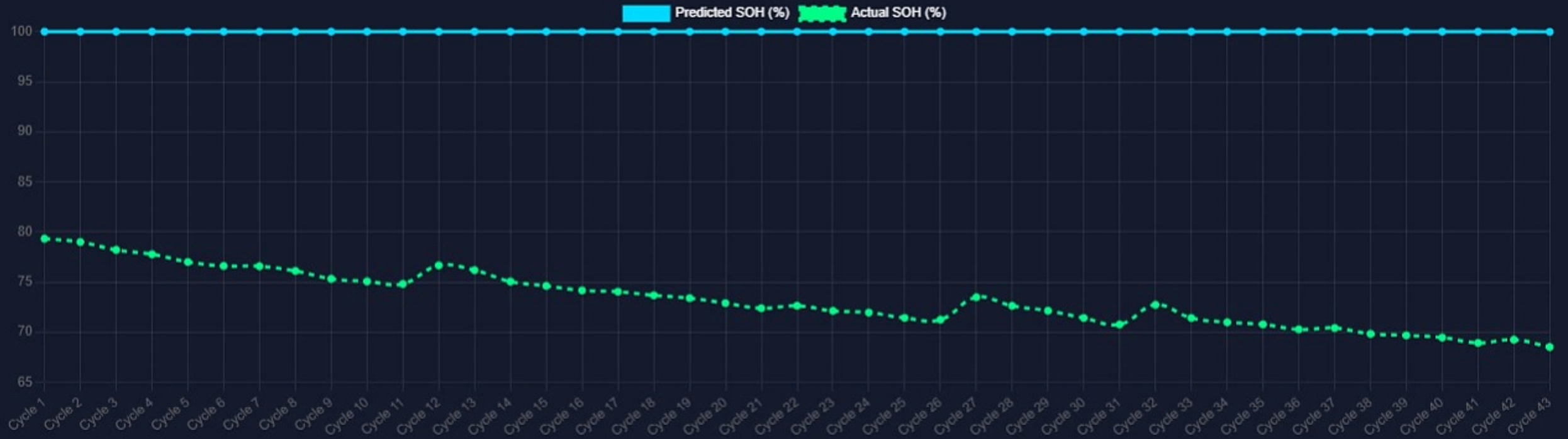
43 / 73

MODEL MAPE

4.09%

BATTERY STATUS

Healthy



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AI POWERED EV BATTERY SYSTEM

INTELLIGENT BATTERY HEALTH DASHBOARD

DARK

LIGHT

SOH STATUS

100.0%

CHARGE LEVEL

40.8%

TEMPERATURE

44.8°C

SYSTEM



ACTIVE



SOH ESTIMATION



RL OPTIMIZATION



MODEL COMPARISON

REINFORCEMENT LEARNING LOGS

Q-Learning (Baseline)



RECOMMENDED ACTION

Moderate

CUMULATIVE REWARD

98.0

ACTION EFFICIENCY

85.0%

Project: On-Time Battery Health Estimation and Optimization in Electric Vehicles

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PROGRESS

55 / 73

MODEL MAPE

4.09%

BATTERY STATUS

Healthy



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AI POWERED EV BATTERY SYSTEM

INTELLIGENT BATTERY HEALTH DASHBOARD

DARK

LIGHT

SOH STATUS

100.0%

CHARGE LEVEL

32.8%

TEMPERATURE

46.8°C

SYSTEM



ACTIVE

SOH ESTIMATION

RL OPTIMIZATION

MODEL COMPARISON

STATE OF HEALTH ESTIMATION

CNN Model



STOP

UPLOAD BATTERY DATA

Click to select a single CSV file

Live SOH Estimation Started...

PROGRESS

59 / 73

MODEL MAPE

4.09%

BATTERY STATUS

Healthy



Predicted SOH (%)



Actual SOH (%)



PROGRESS

73 / 73

MODEL MAPE

4.09%

BATTERY STATUS

Healthy



Project: On-Time Battery Health Estimation and Optimization in Electric Vehicles

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AI POWERED EV BATTERY SYSTEM

INTELLIGENT BATTERY HEALTH DASHBOARD

DARK

LIGHT

SOH STATUS

100.0%

CHARGE LEVEL

21.6%

TEMPERATURE

49.6°C

SYSTEM



PROCESSED

SOH ESTIMATION

RL OPTIMIZATION

MODEL COMPARISON

STATE OF HEALTH ESTIMATION

CNN Model



START SYSTEM

UPLOAD BATTERY DATA

Click to select a single CSV file

Route Finished: All Telemetry Processed.

PROGRESS

73 / 73

MODEL MAPE

4.09%

BATTERY STATUS

Healthy

100

Predicted SOH (%) Actual SOH (%)



AI POWERED EV BATTERY SYSTEM

INTELLIGENT BATTERY HEALTH DASHBOARD

DARK

LIGHT

SOH STATUS

100.0%

CHARGE LEVEL

21.6%

TEMPERATURE

49.6°C

SYSTEM



PROCESSED

SOH ESTIMATION

RL OPTIMIZATION

MODEL COMPARISON

REINFORCEMENT LEARNING LOGS

Q-Learning (Baseline) ▼

RECOMMENDED ACTION

Moderate

CUMULATIVE REWARD

146.0

ACTION EFFICIENCY

85.0%

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